



Software Design Document

ALZ-Foresight

Predicting Alzheimer's Progression Using Multimodal Machine Learning on ADNI Dataset

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Abstract

This project offers a research grade system capable of predicting the Alzheimer's disease (AD) progression based on multimodal machine learning applied to ADNI data. The entire process is outlined in the pipeline that takes in data from structural MRI, clinical text/fields, and cognitive/demographic tables; extracts modality-specific features; and fuses them using robust methods (early/late with missing-modality handling). It generates outputs in two stages: the present AD state and the short-term risk at 12, 24, and 36 months. To provide reliable results, we conduct post-hoc probability calibration and display uncertainty next to reliability diagrams. Grad-CAM is utilized for MRI and feature/attention contributions for non-imaging inputs, ensuring interpretability. A secure web application (React + Node/Express) provides role-based dashboards for clinicians, researchers, and caregivers, and facilitates patient/assessment management, risk visualization, longitudinal tracking, and anonymized exports. The Python ML service is isolated behind a typed API and conservatively records model versions, seeds/ hashes, and artifacts for auditability and reproducibility. Design constraints impose strict adherence to ADNI policy, subject-level data splits to prevent information leakage, and the most efficient compute paths (GPU-preferred, CPU fallback) as a practical aspect. The project has resulted in a calibrated, interpretable, and reproducible end-to-end workflow that integrates various fragmented tools into a single platform that is suitable for education and research (not a medical device) and has a runnable demo as well as documentation aligned to course evaluation criteria.

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1 Introduction

1.1 Purpose

The main objective of this paper is to outline the Software Design Description (SDD) for the “Predicting Alzheimer’s Progression Using Multimodal Machine Learning on the ADNI Dataset” project. This document is intended for all the stakeholders involved in the project, which includes the faculty supervisor, the research team, the software engineers, and the UX designers, and it will describe the methods of the system’s design and construction in a way that satisfies the approved SRS. It covers the system's architecture, principal parts, interfaces, and design choices which assure the application of rigorous methodologies, privacy compliance, and so on for a research/educational (non-medical-device) deployment in terms of interpretability and reproducibility.

1.2 Scope

This SDD covers the design for:

- **Data layer:** ingestion of ADNI-compatible data; subject-level splitting; tracking of provenance.
- **ML layer:** extraction of modality-specific features (MRI, text, tabular); fusion with masks of presence in early/late; prediction in two stages (current AD status and 12/24/36-month risk); calibration of probabilities; generation of explainability artifact (like, for example, Grad-CAM, feature/attention contributions).
- **Application Layer:** web dashboard with secured access (RBAC) for clinicians, researchers, and caregivers; management of patient and clinical records; data representation of risk, uncertainty, longitudinal trends, and recommendations.
- **Integration & Ops:** REST APIs for data/predictions/model info/health; audit/logging; export for research reporting; model versioning and reproducibility assets.

Clinical claims or bedside deployment, on-device acquisition, regulatory submissions (for example, FDA/CE), direct EHR/HL7/FHIR integrations, and additional omics/modalities beyond ADNI (unless explicitly listed as future work) are out of this SDD's scope.

1.3 Definitions & Acronyms

- **ADNI:** Alzheimer’s Disease Neuroimaging Initiative (standardized longitudinal cohort).
- **CN / MCI / AD:** Cognitively Normal, Mild Cognitive Impairment, Alzheimer’s Disease.
- **T1-MRI:** Structural MRI sequence used for volumetry and morphometry.

- **MMSE / CDR-SB / ADAS-Cog:** Common cognitive assessment scales used as tabular inputs.
- **AUC / PR-AUC:** Discrimination metrics for binary classification (ROC and Precision–Recall).
- **ECE / Brier Score:** Calibration metrics, quantify alignment between predicted probabilities and observed outcomes.
- **Early vs. Late Fusion:** Feature-level concatenation vs. ensembling/head-level aggregation across modalities.
- **Modality Dropout / Presence Mask:** Training and runtime strategies to remain robust when one or more modalities are missing.
- **Grad-CAM / SHAP / Attention:** Explanation methods for imaging, tabular, and text inputs, respectively.
- **Subject-level Split:** Ensures all visits of a subject reside in a single fold to prevent temporal/data leakage.

2 System Overview

2.1 Product Functions

1. **Data Curation:** Import ADNI data, enforce subject-level splits, manage site/scanner/protocol metadata, optionally perform harmonization.
2. **Feature Extraction:** Encoders for MRI (CNN/TL), clinical text (transformer or structured fields), and tabular (scaled cognitive/demographic features).
3. **Fusion & Inference:** Early vs. late fusion pipelines, modality dropout for missing data.
4. **Two-Stage Heads:** (a) Current AD at baseline, (b) probability of AD within 12–36 months.
5. **Calibration & Explainability:** Temperature/Platt scaling, reliability diagrams, Grad-CAM/attention maps.
6. **Dashboard:** Patient-level views (risk gauges and uncertainty bands), cohort comparisons, longitudinal trends, and actionable recommendations.
7. **User & Patient Management:** Auth, roles, CRUD for patients, assessments, medications, activities, and appointments.
8. **APIs:** REST endpoints for data, predictions, and model info/health.
9. **Audit & Logging:** User actions, data access, and model inference logs.

2.2 Assumptions & Dependencies

1. ADNI Data Use Agreement is in force, only permitted de-identified fields are stored/exported.
2. Availability of pretrained encoders (medical CNNs/transformers) to bootstrap performance.
3. Compute capacity supports 3D MRI preprocessing/inference, CPU path remains responsive for tabular/text.
4. Stack continuity: React/Node/Mongo frontend/backend and Python (Flask/FastAPI) ML service with pinned versions.

2.3 User Characteristics

1. **Clinician (primary):** Reviews patient risk & explanations, enters assessments, sets reminders.

2. **Caregiver/Patient (secondary):** Views simplified risk insights & recommendations.
3. **Admin/Researcher:** Manages users/roles, model versions, data subsets.

2.4 User Problem Statement

1. The current tools being utilized in this specific medical field are scattered and frequently out of sync, exhibiting a deficiency in unified multimodal fusion and clear near-term predictions.
2. Medical professionals demand a strongly dependable platform that integrates imaging, cognitive scores, and clinical data along with validation being made obvious.

2.5 User Objectives

1. Reliable now/near-term predictions with leakage-free methodology and demonstrated calibration.
2. Interpretable outputs (saliency/contributions) that are communicable to patients and actionable in care planning.
3. Transparent metrics, ablations, and model cards, fast, coherent dashboard UX.

2.6 General Constraints

1. Research-only software (not a medical device), disclaimers shown in UI and reports.
2. Strict adherence to ADNI policy: de-identification, no re-identification or third-party redistribution, auditable access.
3. Compute/runtime limits: cache embeddings, defer heavy MRI explanations to background jobs when needed.
4. Reproducibility: subject-level CV, frozen split manifests, fixed seeds/configs, stored preprocessing hashes, encoder/calibration versions

3 System Requirements Specification

3.1 Functional Requirements

3.1.1 Data & Modeling

Functional Requirement (*FR-01.1*)

Priority & Ranking: System Depends On It (#1)

Function Name	Data Ingestion, Harmonization & Subject-Level Cohorting
Description	Allows the system to implement a reproducible pipeline to import ADNI multimodal data and produce a leakage-safe, subject-level research dataset with explicit train/validation/test splits.
Critically	All downstream feature extraction, fusion, and modeling depend on a correct, leakage-safe dataset.
Technical issues	Anticipated challenges include heterogeneous file formats and identifiers across ADNI phases, visit time alignment, site/scanner/protocol variability, incomplete modalities, and the risk of leakage if scalers/imputers are fit outside the training partition. Throughput and resumability are required for large I/O, checksums and retry logic are necessary to guarantee integrity.
Cost and schedule	3–5 engineering days for ingestion, QC, harmonization, and split generation, and 2 days for validation, documentation, and provenance checks.
Risks	The primary risks are hidden data leakage (subject overlap or future information bleeding into training), site/scanner bias that shifts distribution, label noise when deriving progression windows, and silent record loss that distorts cohort characteristics.
Dependencies with other requirements	This requirement enables FR-01.2 (Feature Extraction), FR-01.3 (Fusion & Missing-Data Handling), FR-01.4 (Two-Stage Prediction), FR-01.5 (Calibration & Interpretability), and FR-05.2 (Model Versioning/registry).
Pre-Condition	Valid ADNI credentials & local data dictionary.
Post-Condition	Cleaned, aligned per-subject dataset with split manifests.
Inputs	Raw ADNI files & CSVs, and configuration for inclusion/exclusion.
Outputs	Standardized parquet/NPZ/JSON, split files and provenance log.

Functional Requirement (FR-01.2)**Priority & Ranking:** System Depends On It (#1)

Function Name	Multimodal Feature Encoders
Description	Allows the system to convert each modality into a fixed-length embedding using stable, reproducible pipelines. MRI volumes are preprocessed and encoded with a 3D CNN or transfer-learning encoder, clinical text or structured fields are normalized and embedded with a transformer or MLP, tabular cognitive and demographic variables are imputed and scaled before MLP encoding. All embeddings carry provenance (subject, visit, preprocessing and encoder versions) to guarantee traceability.
Critically	Downstream fusion and prediction depend on embeddings.
Technical issues	MRI memory/VRAM footprint, heterogeneous field-of-view, text length truncation, leakage from future visits, imputation bias, ensuring subject-level split integrity during scaler/fit steps.
Cost and schedule	3–5 engineering days to stand up pipelines if using existing libs, and 2–3 days for QA and caching.
Risks	Collinearity across tabular features, site/protocol bias in MRI, missing modalities at inference, encoder version drift.
Dependencies with other requirements	FR-01.1 (ingestion), FR-01.3 (fusion), FR-01.5 (calibration needs stable features), FR-05.2 (versioning of encoders).
Pre-Condition	Cleaned inputs available, split manifests frozen, encoder configs chosen.
Post-Condition	Modality-specific tensors cached to disk/DB, with metadata and checksums.
Inputs	T1-MRI volumes, clinical text/structured CSVs, and cognitive/demographic tables, preprocessing configs.
Outputs	MRI embedding, text embedding (or structured), embedded tables and metadata JSON.

Functional Requirement (FR-01.3)**Priority & Ranking:** System Depends On It (#1)

Function Name	Multimodal Fusion Layer
Description	Allows the system to combine modality embeddings using either early fusion (feature concatenation) or late fusion (ensembling of

	per-modality heads). The fusion layer must operate when one or more modalities are absent by learning presence-aware gating and training with modality dropout. When site or scanner shifts are present, optional domain alignment reduces bias while preserving predictive utility.
Critically	Enables inference when some modalities are missing and improves performance over unimodal baselines.
Technical issues	Training instability with adversarial loss, imbalance in modality quality, ensuring gradients flow when modality is absent, calibration drift across fusion types.
Cost and schedule	3–4 days to implement and benchmark early vs late fusion, and 2 days for missing-data ablations.
Risks	Overfitting to sites, degraded performance if gating mislearned, silent failures when a modality path breaks.
Dependencies with other requirements	FR-01.2 (embeddings), FR-01.5 (post-fusion calibration), FR-05.2 (model variants registry).
Pre-Condition	Valid embeddings for at least one modality, presence mask available.
Post-Condition	Unified fused representation or ensemble probability with contribution report.
Inputs	We optionally compute MRI, text, and tabular feature vectors, record which modalities are present using a mask, and apply a fusion configuration to combine them.
Outputs	Combined feature vector (early) or combined prediction probabilities (late), contribution weights.

Functional Requirement (**FR-01.4**)

Priority & Ranking: Extremely High (#2)

Function Name	Prognostic Heads (Now & 12–36m)
Description	Allows the system to produce a calibrated probability for current AD status at baseline and separate horizon-specific probabilities for progression at 12, 24, and 36 months. Labels are constructed to avoid temporal leakage and account for censoring or class imbalance. The module may optionally expose uncertainty estimates to communicate confidence.
Critically	Central research output communicated to users.

Technical issues	Label curation for progression, censoring, class imbalance, temporal leakage, aligning visit times.
Cost and schedule	4–6 days to implement/train/evaluate, and 2 days for uncertainty estimation.
Risks	Over-confident predictions, confusing multi-horizon outputs, and degradation under missing MRI.
Dependencies with other requirements	Over-confident predictions, confusing multi-horizon outputs, and degradation under missing MRI.
Pre-Condition	Trained fusion encoder, and labeled datasets with subject-level splits.
Post-Condition	Saved model weights, eval metrics, and serving endpoints.
Inputs	Combined feature vector or per-modality probabilities, label manifests, and training configs.
Outputs	Calibrated probability of Alzheimer’s now, probability of it in 1 to 3 years and uncertainty.

Functional Requirement (**FR-01.5**)

Priority & Ranking: Extremely High (#2)

Function Name	Calibrator & XAI Artifacts
Description	Allows the system to calibrate predicted probabilities on a held-out split and report reliability with ECE/Brier and reliability diagrams. Generate patient-level explanations appropriate to each modality (e.g., Grad-CAM for MRI, SHAP for tabular, attention/IG for text). Explanations are packaged with model/version metadata for consistent rendering in the UI.
Critically	Mandatory for trustworthy probabilities and educational value.
Technical issues	Calibration shift across sites, saliency robustness, SHAP cost on large tabular sets, linking artifacts to visits securely.
Cost and schedule	2–3 days for calibration and plots, 3–4 days for XAI artifacts and UI hooks.
Risks	Misinterpretation of saliency, leakage if explanations expose sensitive text, compute overhead.
Dependencies with other requirements	FR-01.4 (raw probabilities), FR-03.2 (explainability views), and FR-05.2 (store calibrator version).
Pre-Condition	Trained model, reserved validation split.
Post-Condition	Calibrated scorer, reliability/interpretability artifacts persisted.

Inputs	Model logits/probabilities, validation labels, MRI volumes, tabular/text inputs.
Outputs	Calibrated probabilities, ECE/Brier, Grad-CAMs, SHAP/attention visuals, and per-patient explanation bundle.

3.1.2 Web Application & APIs

Functional Requirement (**FR-02.1**)

Priority & Ranking: Extremely High (#2)

Function Name	RBAC Auth Gateway
Description	Allows the system to provide secure login, token issuance, and role-based access control for all protected resources. Sessions rely on short-lived access tokens with refresh support and middleware guards on server routes. The system logs claims and key events to support auditing and incident response.
Critically	Protects patient data and model endpoints.
Technical issues	Token revocation/rotation, secure storage of refresh tokens, clock skew, CORS and CSRF.
Cost and schedule	2–3 days to implement, 1 day for penetration checks.
Risks	Credential stuffing, token leakage, misconfigured CORS exposing APIs.
Dependencies with other requirements	FR-05.1 (audit actions), and NFR-Security.
Pre-Condition	User exists or Admin can invite, HTTPS configured.
Post-Condition	Authenticated session with role claims, denied access otherwise.
Inputs	Email/password (or SSO, if configured), and JWT.
Outputs	Access/refresh tokens, and standardized auth errors.

Functional Requirement (**FR-02.2**)

Priority & Ranking: Extremely High (#2)

Function Name	Clinical CRUD Services
Description	Allows the system to persist and manage core clinical entities (Patients, Assessments, Medications, Activities, Appointments) with validation and indexing. Provide search, pagination, and filters, enforcing ownership and permissions at the API layer. Emit structured events for audit trails and downstream analytics.

Critically	Core clinical data backbone for predictions and dashboards.
Technical issues	Referential integrity, concurrent edits, time-zone handling, validation of medical units.
Cost and schedule	4–6 days including UI forms and server validators.
Risks	Data inconsistency, accidental PII exposure, orphaned records if cascading deletes not handled.
Dependencies with other requirements	FR-02.1 (auth), FR-03.1 (dashboard consumes), and FR-05.1 (audit).
Pre-Condition	Authenticated user with proper role.
Post-Condition	Persisted entities, index updates, and events logged.
Inputs	Form payloads (patient demographics, assessment scores, meds, activities, and appointments).
Outputs	Saved documents, list/search results, and validation errors.

Functional Requirement (**FR-02.3**)

Priority & Ranking: Extremely High (#2)

Function Name	Inference Proxy & Health
Description	Allows the system to expose normalized endpoints that validate input, forward inference requests to the Python ML service, and standardize responses (probabilities, uncertainty, explanations). Implement timeouts, retries, and health checks to keep the UI responsive and inform users about model readiness. Include model/version information in responses for traceability.
Critically	Connects UI/DB with ML service reliably.
Technical issues	Cross-service timeouts, serialization of large artifacts (Grad-CAM images), version skew.
Cost and schedule	2–3 days including error handling and tests.
Risks	Silent degradation if fallback paths not explicit, DOS via heavy inference requests.
Dependencies with other requirements	FR-01.4/01.5 (model outputs), and FR-05.3 (health/alerts).
Pre-Condition	ML API reachable, service discovery configured.
Post-Condition	Standardized inference responses for UI/storage.
Inputs	Patient/visit payload, requested horizons, and flags for explanations.

Outputs	Probability of Alzheimer's at a specific time, uncertainty, explanation bundle, and model metadata.
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3.1.3 Dashboard & Visualization

Functional Requirement (*FR-03.1*)

Priority & Ranking: High (#3)

Function Name	Risk & Longitudinal UI
Description	Allows the system to present a cohesive patient view with risk gauges, uncertainty bands, horizon selectors, and longitudinal charts of assessments, medications, activities, and appointments. Surface quick actions and clear status indicators to streamline clinical workflows. All visuals reflect the latest data and selected model version.
Critically	Primary decision surface for clinicians.
Technical issues	Chart performance, syncing time axes, accessibility, caching.
Cost and schedule	4–6 days for views and 2 days for polishing.
Risks	Misinterpretation of risk visuals, stale caches, slow charts on large histories.
Dependencies with other requirements	FR-02.2 (data), FR-02.3 (predictions), and FR-01.5 (artifacts).
Pre-Condition	Authenticated clinician, patient selected with data.
Post-Condition	Up-to-date rendered dashboard panels.
Inputs	Patient entities, prediction results, and calibration/XAI artifacts.
Outputs	Interactive UI components, and exportable charts.

Functional Requirement (*FR-03.2*)

Priority & Ranking: High (#3)

Function Name	XAI UI Panels
Description	Allows the system to render modality-specific explanation artifacts and concise narratives that help users interpret each prediction. Visual elements follow consistent scales and legends and are linked to the patient visit and model version. Secure download options are provided for research reporting when authorized.
Critically	Increases transparency and educational value.

Technical issues	Serving large images, PHI safety in text snippets, consistent color scales.
Cost and schedule	3–5 days including accessibility and downloads.
Risks	Over-trust due to visual artifacts, data leakage via text tokens.
Dependencies with other requirements	FR-01.5 (artifacts), FR-02.3 (delivery), and NFR-Privacy.
Pre-Condition	Prediction with explanations available.
Post-Condition	XAI panels rendered, artifacts linked and/or downloaded.
Inputs	Grad-CAM images, SHAP rankings, attention snippets, model metadata.
Outputs	Explanations UI, and downloadable assets.

3.1.4 Audit & Operations

Functional Requirement (*FR-04.1*)

Priority & Ranking: High (#3)

Function Name	Audit Logger
Description	Allows the system to record security-relevant and clinical events with timestamps, actor identifiers, patient context, and outcomes. Logs are immutable, queryable, and exportable with defined retention periods. Correlation identifiers enable tracing requests across services.
Critically	Forensic traceability and compliance.
Technical issues	Log volume, PII minimization, secure storage, correlation IDs.
Cost and schedule	2–3 days to implement, and 1 day for exports/indexes.
Risks	Storage bloat, missing events, insecure log access.
Dependencies with other requirements	FR-02.1 (user identity), and NFR-Security.
Pre-Condition	Authenticated routes, log sink configured.
Post-Condition	Durable, queryable audit records.
Inputs	Event objects from middleware/services.
Outputs	Log streams, exports, and alerts (via FR-05.3).

Functional Requirement (*FR-04.2*)

Priority & Ranking: High (#3)

Function Name	Model Registry
Description	Allows the system to maintain a registry of model artifacts, training data hashes, metrics, and calibration parameters. Allow controlled promotion and rollback of serving versions with auditable change logs. Expose lightweight APIs so the UI and backend can resolve the active model consistently.
Critically	Reproducibility and safe iteration.
Technical issues	Artifact storage, schema evolution, hash reproducibility.
Cost and schedule	2–3 days to set up, and 1 day for UI hooks.
Risks	Serving wrong version, orphaned artifacts, metric mismatch.
Dependencies with other requirements	FR-01.2/01.4/01.5, and FR-02.3 (serving needs active model).
Pre-Condition	Trained models exist, storage configured.
Post-Condition	Versioned entries, one marked active.
Inputs	Model files, metrics JSON, and notes.
Outputs	Registry records, and active model pointer.

Functional Requirement (**FR-04.3**)

Priority & Ranking: High (#3)

Function Name	Resilience & Health Monitor
Description	Allows the system to standardize error envelopes, implement timeouts and retries, and expose health/readiness probes for all services. Trigger actionable alerts on repeated failures or unhealthy states while minimizing noise. Provide user-facing status messages that explain degraded behavior without exposing internals.
Critically	Operational stability.
Technical issues	False positives, noisy alerts, cascading timeouts.
Cost and schedule	2–3 days including dashboards for health.
Risks	Alert fatigue, hidden partial failures.
Dependencies with other requirements	FR-02.3 (proxy), and NFR-Reliability.
Pre-Condition	Deployed services with metrics endpoints.
Post-Condition	Measurable health, actionable alerts, graceful errors.
Inputs	Error events, health pings, thresholds.

Outputs	Alert notifications, error responses, and status pages.
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Functional Requirement (*FR-04.4*)

Priority & Ranking: Moderate (#4)

Function Name	Research Exporter
Description	Allows the system to generate anonymized export bundles that include prediction tables, logs, and selected visualizations for approved cohorts and time windows. Ensure de-identification rules are enforced consistently before packaging. Provide progress feedback and an auditable record of exports.
Critically	Convenience for research dissemination.
Technical issues	Large export bundles, chart rendering server-side, privacy checks.
Cost and schedule	2–3 days.
Risks	Accidental re-identification, incomplete exports.
Dependencies with other requirements	FR-03.1/03.2 (visuals), FR-05.1 (logs), and NFR-Privacy.
Pre-Condition	Data present and user authorized.
Post-Condition	Downloadable archive generated.
Inputs	Query params, selection of artifacts.
Outputs	ZIP with CSV/JSON and images.

3.1.5 Compliance & Ethics

Functional Requirement (*FR-05.1*)

Priority & Ranking: Moderate (#4)

Function Name	Consent & Policy Guardrails
Description	Allows the system to enforce ADNI data-use rules by tagging sensitive fields, validating requests against consent flags, and blocking or redacting disallowed operations. Administrators can configure additional research-only restrictions for exports and displays. Every enforcement action is logged for audit.
Critically	Ethical and legal compliance.
Technical issues	Mapping ADNI policy to fields, preventing side-channel re-identification, policy drift.
Cost and schedule	2 days.

Risks	Non-compliance, accidental PHI exposure.
Dependencies with other requirements	FR-02.2 (data fields), FR-05.4 (export), and NFR-Privacy.
Pre-Condition	Policy documented, fields tagged with sensitivity.
Post-Condition	Requests violating policy are blocked or redacted.
Inputs	User role, requested fields/actions.
Outputs	Allowed action or policy error, redacted payloads.

Functional Requirement (*FR-05.2*)

Priority & Ranking: Low (#5)

Function Name	Risk Education Module
Description	Allows the system to provide inline explanations of probabilities, calibration, uncertainty, and thresholding with concise, plain-language help. Content appears contextually in the dashboard and links to a glossary and a short “how to read this” guide. Materials are versioned alongside the active model.
Critically	Reduces misuse and improves decisions.
Technical issues	Localization/readability, keeping content synced with model updates
Cost and schedule	1–2 days.
Risks	Over-simplification or misunderstanding, outdated content after model change.
Dependencies with other requirements	FR-03.1/03.2 (UI), and FR-01.5 (calibration terms).
Pre-Condition	Predictions visible, user in dashboard.
Post-Condition	Help panels/tooltips rendered.
Inputs	Current model metadata, and glossary content.
Outputs	Contextual help and disclaimers.

3.2 Non-Functional Requirements

3.2.1 Security (*NF-R1*) | Priority: High

The entire clinical research data and model artifacts need to be comprehensively protected by the system from the storage through to the execution. All APIs are secured by the TLS

protocol, authentication is accomplished through the use of short-lived access tokens that can be refreshed, and authorization is done by applying the least-privilege role scopes that consist of Admin, Clinician, Researcher, Caregiver/Patient at the route and query levels. Secrets are kept in separate stores for different environments, passwords are protected through the use of salts and hashing, all inputs are checked for validity on the server side in order to avoid risks of injection and deserialization, and there is private object storage that uses signed URLs for granting access to the MRI saliency maps and reports. Logs do not contain PHI and are divided according to tenant/project. The threat model takes into consideration attacks such as credential stuffing, token leakage, CSRF/CORS misconfiguration, model endpoint abuse, and artifact tampering, the mitigations include rate limiting, IP throttling on auth routes, strict CORS, token rotation, audit trails, and checksums for integrity on model and embedding files. Security verification activities include static analysis, dependency scanning, and periodic penetration testing of the authentication gateway and machine learning endpoints.

3.2.2 Reliability & Recovery (NF-R2) | Priority: High

In case of issues with the ML service or a downstream datastore, clinical workflows must not be interrupted. The backend applies timeouts, retries with exponential backoff, and circuit breakers about the Python ML API, when the UI is degraded, it shows a clear state and allows to load previously cached results while queuing non-critical jobs. Health and readiness probes input a simple operations dashboard, MongoDB and the model registry backups are done daily with point-in-time recovery, artifact stores are versioned and non-writable. The service level expectations are 99.5% monthly availability for the web/API tier and a successful recovery of the primary database within four hours from a severity-1 incident. Fault injection tests verify that the system still allows reads, blocks unsafe writes, and automatically recovers from typical failure types such as ML timeouts, network partitions, and storage hiccups.

3.2.3 Maintainability (NF-R3) | Priority: Medium/High

The architecture of the codebase is separated into packages for frontend, backend and ML services with defined interfaces, semantic versioning and continuous integration for unit and integration tests that are automated. The common agreements (schemas used for prediction, calibration metadata) are created just once and then shared among the services to keep drift under control. Linting, formatting and typed models contribute to lowering the complexity that is not intended, the infrastructure is coded to prevent configuration discrepancies between environments. Operational activities like model promotion, rollback,

and calibrator updates are written down and a change log tracks the evolution of the schema and endpoints. So the target is that a new employee can carry out a minor implementation or bug fix in one day, also, a routine update does not necessitate a coordinated downtime that is prearranged. Maintenance is measured in terms of test coverage, build health, and mean time to restore after regressions.

3.2.4 Portability (NF-R4) | Priority: Medium

Deployment must be done in a way that it can be used in both on-premises and cloud environments without any code alterations. Services are deployed in containers and are configured through environment variables, the use of external databases, caches, and storage is done through drivers so that MongoDB, Redis, and S3-compatible storage can be easily replaced with their managed versions when they become available. In the machine learning service, GPU usage is determined by the current availability: if there is no CUDA, the system will use CPU for tabular/text paths and will wait for an MRI-heavy explainability job to be processed in order not to fail. Build artifacts are created for multiple architectures if it is feasible, and the application can work with standard ingress and logging practices for local or managed stack connectivity. Portability is confirmed by launching the entire system with docker-compose on a local machine and then using the same images on a cloud runner with only changing the environment variables.

3.2.5 Extensibility (NF-R5) | Priority: Medium

It should be possible to incorporate new modalities, features, and model variants into the system without carrying out disruptive refactors. Encoders are designed to adhere to a standard interface that produces both embeddings and provenance, which in turn enables the fusion layer to receive more inputs like PET or CSF biomarkers if they are present. The model registry retains any kind of metadata that is needed so that different architectures can be run at the same time and A/B tested. The dashboard on the UI side is made up of panels that are connected to a unified prediction schema, thus allowing the addition of new horizons, uncertainty measures, or explanation types without having to rewrite existing views. The system shows its extensibility through the incorporation of one new feature family or encoder within a single sprint, followed by complete propagation to inference, calibration, storage, and visualization.

3.2.6 Accessibility & Reusability (NF-R6) | Priority: Low

The authorized contributors should have accessibility to the system's code along with the data assets and make them available for research replication. Every dataset has its respective split manifests, preprocessing hashes, and data dictionaries in order to enable the exact replication of experiments; models consist of training configs, metric cards, and calibration parameters. The reusability of UI components (risk gauges, longitudinal charts, explanation panels) is an advantage as they are made available internally in libraries and thus, across the pages they are reused. The documentation addresses environment setup, experiment reproduction, and data-use compliance, which allows new researchers to contribute responsibly as it lowers the barrier for them. Accessibility is evaluated based on the time taken to reproduce a benchmark experiment starting from a clean checkout and the ratio of components that are shared across features instead of being rewritten.

3.2.7 Resource Utilization (NF-R7) | Priority: High

The whole process of inference and explanation should be so efficient that the feedback given to clinicians would be quick and the compute costs would still be predictable. The completion time for tabular/text-only requests should be approximately three seconds on CPU, for MRI-based requests, the expectation is to stay within fifteen seconds when Grad-CAM is asked for, with asynchronous rendering in case it is necessary. Caching of embeddings, deduplication of the repeated requests within a very short period, and streaming of large artifacts from object storage instead of through the API are among the measures taken to reduce resource usage. The batch prediction endpoints are made in such a way that they can support cohort analyses without the risk of overloading interactive services, and the background workers are scaled apart from the web tier to avoid noisy-neighbor effects. Tracking of resource usage is done per request type and per model version so that the hotspots can be either tuned or offloaded, also, the budgets are enforced through queue limits and concurrency caps.

3.2.8 UI Accessibility & Serviceability (NF-R8) | Priority: Medium/High

The interface has to be easy to use and to maintain in actual clinical environments. The pages have been designed following the WCAG 2.1 AA principles and they provide keyboard navigation, visibility of focus, and sufficient contrast between the text and background along with proper labeling of charts and icons, risk and uncertainty are communicated in straightforward terms with illustrations to avoid misunderstanding. The UI is still usable on small screens and with low-bandwidth connections, and it offers progressive loading and

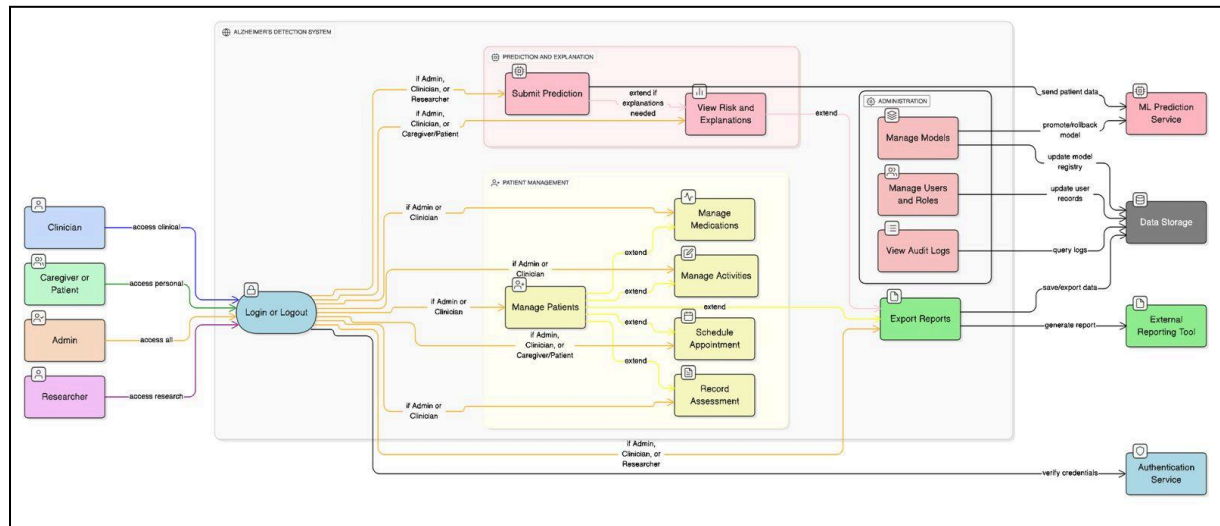
visible empty states. Features for serviceability include a machine learning availability status indicator in the app, error IDs that can be copied and correspond to the server logs, and a channel for clinicians to send feedback about patient-specific issues and predictions. Accessibility is assured through the combination of automated checks, and user testing while serviceability is assessed by the difficulty of linking a user-reported problem to a log trace and resolution steps.

3.2.9 User Activity Logging (NF-R9) | Priority: High

All interactions that are significant from a security or clinical perspective are logged in such a way as to create an auditable trail that can be defended, without revealing any PHI. The system logs all the mentioned events, like authentication, CRUD operations, data exports, and also timestamps, and identifies the actors, the patient context, the route names, request fingerprints, and outcomes along with their sensitive payloads, which are summarized instead of being stored. The correlation IDs trace a request throughout its journey so that the investigators can map the flows across different services, and the retention policies make sure that the logs stay accessible for the purpose of review and at the same time the data-use agreements are observed. The access to the logs is limited to specific roles and not detectable by tampering, operational analytics also signal potential misuse or faults in the system by receiving the rates and anomalies. The completeness of the audit is tested from time to time using simulated user journeys and by checking the UI actions against the events recorded.

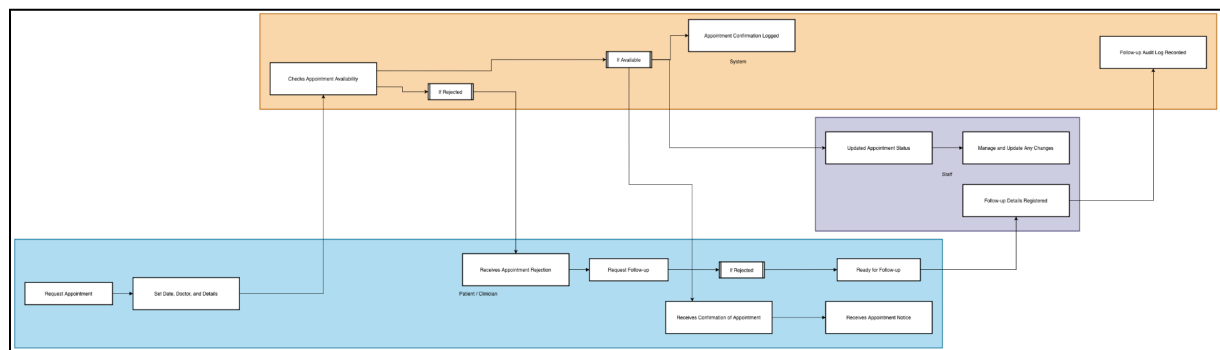
3.3 Use Case Diagrams and Scenarios

3.3.1 Use Case Diagram

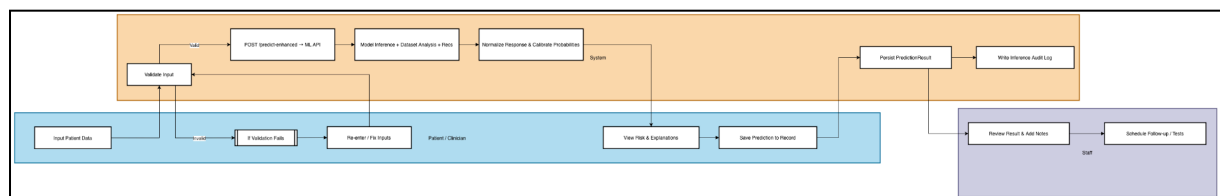


3.3.2 Activity Diagram

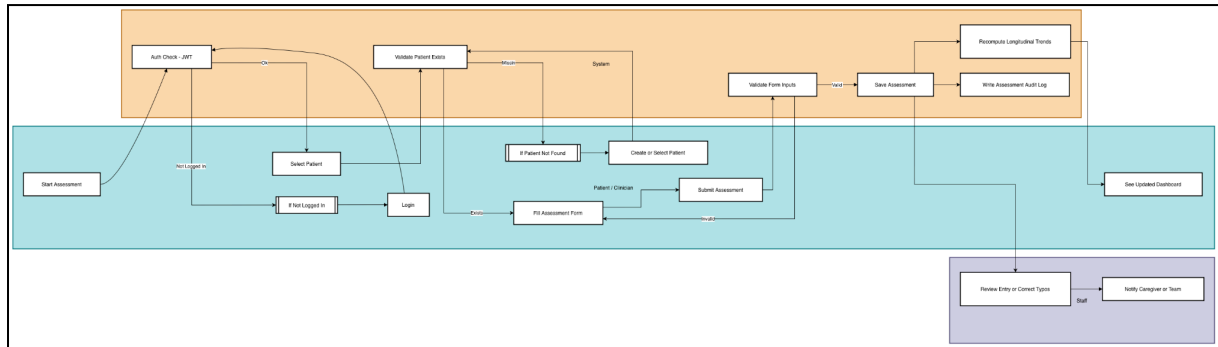
1) Schedule Appointment (With Accept/Reject and Follow-up)



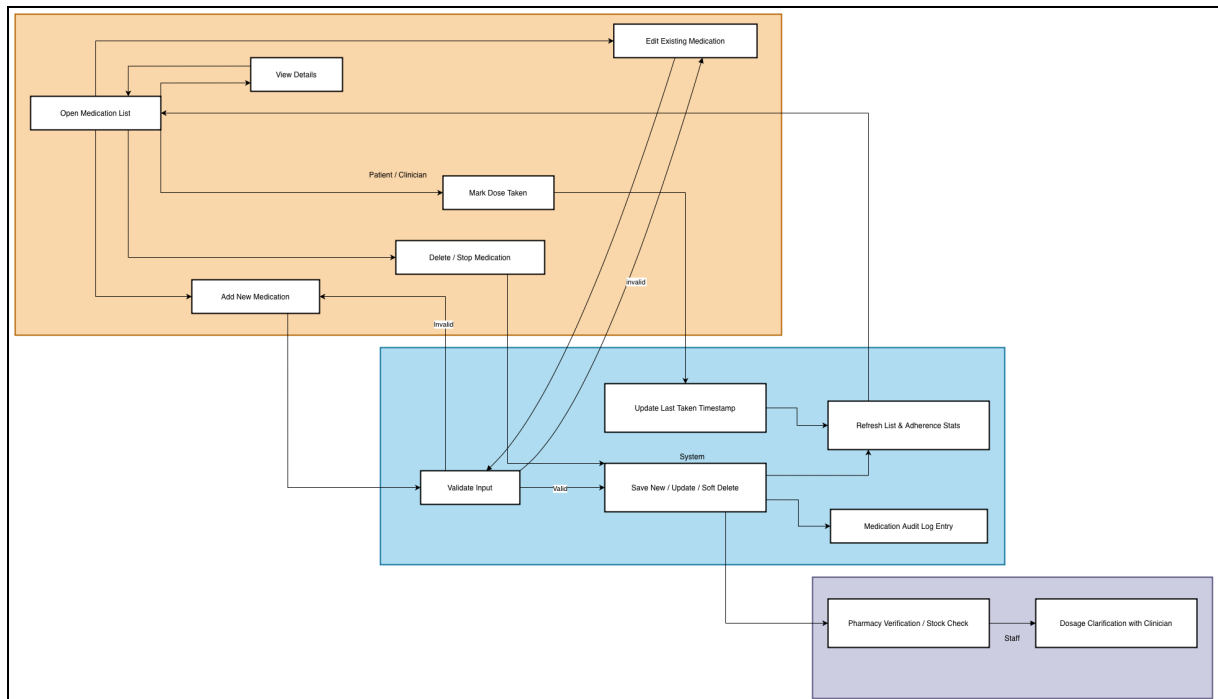
2) End-to-End AI Prediction



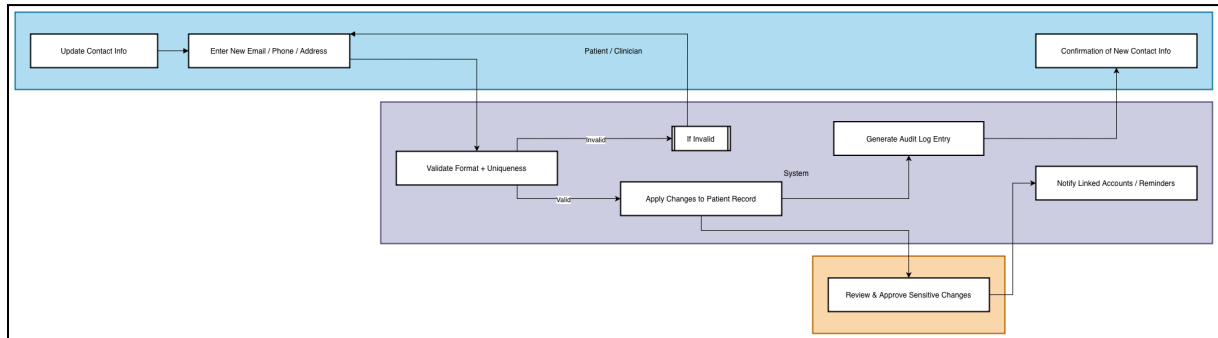
3) Record Assessment (With Authentication and Patient Existence)



4) Medication Management (Add / Edit / Delete / Mark Taken)

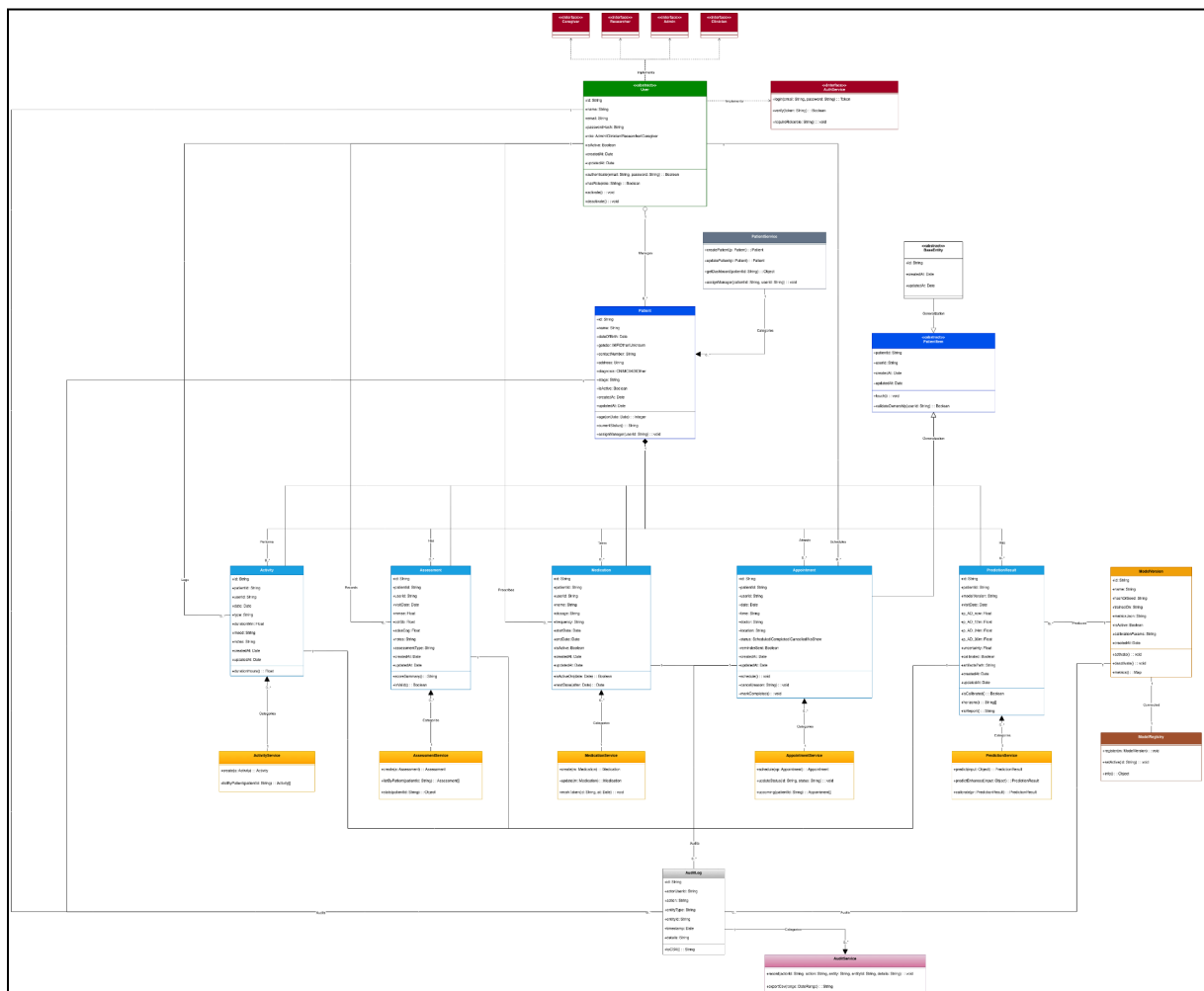


5) Contact Information Update (Validate and Audit)



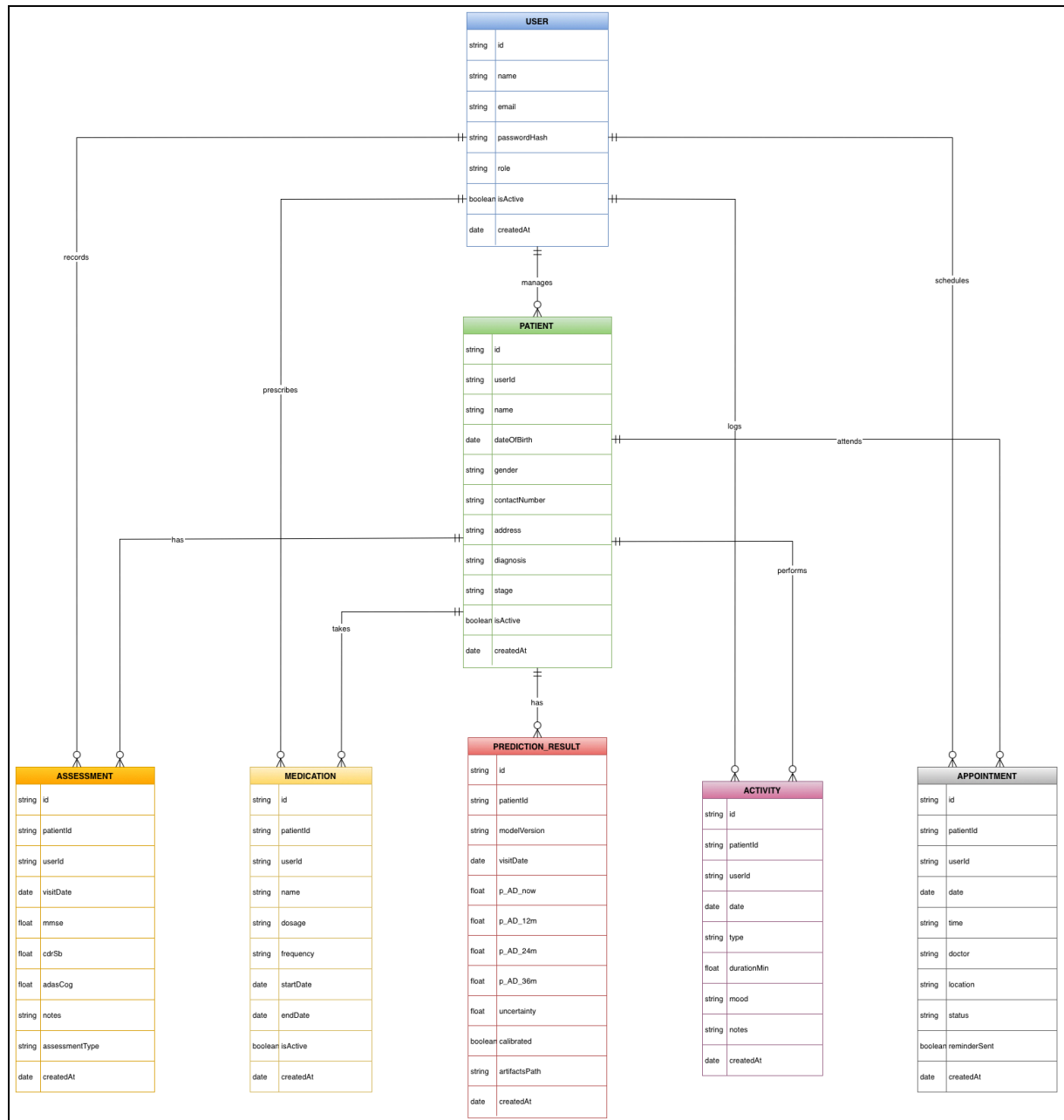
3.3.3 Class Diagram

- **Entities:** User, Patient, Assessment, Medication, Activity, Appointment and PredictionResult.



3.3.4 Entity Relationship Diagram

- Collections/Tables reflecting the domain entities with one-to-many relationships from User and Patient.



4 Architectural Design

4.1 Architecture Overview

Our system uses a client-server model and a three-tier architecture with a services-by-concern split (Web UI, REST API, ML inference) for containerized deployment. The Web UI (React) communicates solely with the Backend API (Node/Express), which manages data access (MongoDB), requests the Python ML Service (Flask/FastAPI) for inference/calibration, and logs audit and model version metadata. The services are stateless (horizontal scalability), have health/readiness endpoints, and require JWT-based RBAC at the API gateway. MongoDB holds the persistent state (clinical data, predictions, audit), while heavy artifacts (e.g., Grad-CAM images) are stored as files or objects and referenced by their paths.

Main Technology Decisions:

- **Standalone ML Service:** It keeps Python/PyTorch dependencies apart, gives API the feature of being light, and allows GPU nodes only where necessary.
- **Stateless Web/API:** It allows easy horizontal scaling and blue-green deployments; the whole state is in MongoDB or artifact storage.
- **RBAC & Audit Everywhere:** JWT with short-lived tokens and irreversible audit trails meet the requirements for privacy and research traceability.
- **Calibration & XAI as first-class:** The post-hoc calibrators and artifact generation keep the probabilities trustworthy and interpretable as they are a part of the serving path.
- **Reproducibility Hooks:** The model registry entries include hashes/seeds, split manifests are locked, and the provenance is kept with predictions.
- **Containerized Delivery:** The same compose stack operates locally, in staging, or on a single VM with minimal env drift.

4.2 Technology Stack

Frontend (Web UI)

- React (Vite or CRA), React Router, axios/fetch respectively.
- Recharts/Chart.js for risk/longitudinal views (like charting).
- Authentication through JWT properly secured (HTTP-only cookie or bearer),

role-aware routing.

- A11y: Components and keyboard navigation that comply with WCAG standards.

Backend API (Application Tier)

- Node.js and Express were utilized in building the REST API.
- Next to it were authentication and role-based access control middleware, validating input, limiting requests, and making security enhancements for CORS and CSRF.
- Data was processed through Mongoose Object Data Modeling (ODM) to the MongoDB database for patients, assessments, medications, activities, appointments, prediction results, audit logs, and model versions.
- The machine learning proxy was set for the Python service with timeouts and retries along with health and readiness endpoints.
- All events (actor, entity, action, outcome, correlation IDs) were logged to audit.

ML Service (Inference & Explainability)

- **Python** (Flask or FastAPI).
- **PyTorch / scikit-learn** for encoders, fusion (early/late), heads (now & 12/24/36m).
- **Calibration** (temperature/Platt), **metrics** (AUC/PR-AUC, ECE, Brier).
- **Explainability**: Grad-CAM for MRI, SHAP/attention for tabular/text; artifacts written to storage and referenced by API.
- Model registry integration (active version, hashes/seeds, calibration params).

Data & Storage

- **MongoDB** (≥ 5.0) as primary store; patient-centric indexing (patientId, visitDate, createdAt).
- File/object storage for large artifacts (e.g., Grad-CAM PNGs), referenced via secure URLs.
- Optional Redis cache for sessions/hot queries (deployment-dependent).

Deployment & Ops

- **Docker / Docker Compose** for local & single-VM deployments.
- **Health probes** on API and ML; logs to stdout (JSON).
- Backups for Mongo; model artifacts versioned and immutable.
- Optional **NGINX** as reverse proxy/TLS terminator in production.

5 Component Design

5.1 Modules and Components

The architecture of the system comprises three principal logical layers: the Web User Interface, the Backend API, and the Machine Learning Service. Each layer consists of dedicated modules, whose responsibilities are clearly outlined.

5.1.1 Web User Interface Modules

- **Authentication Module:** This module is responsible for the management of user logins, logouts, and sessions via JWT-based authentication. It applies a role-based access control strategy for the various users such as clinicians, caregivers, researchers, and administrators.
- **Patient Dashboard Module:** This module is responsible for the visualization of patient profiles, the display of longitudinal clinical data, the representation of risk predictions, the indication of uncertainty bands, and the drawing of progression timelines.
- **Clinical Management Module:** This module is responsible for the secure API calls and allows performing CRUD operations for patients, assessments, medications, activities, and appointments.
- **Prediction Visualization Module:** This module is responsible for the presentation of AI prediction results that include current and future risk (12/24/36 months), calibration indicators, and explainability artifacts such as Grad-CAM and feature contributions.

5.1.2 Backend API Modules

- **Auth & RBAC Module:** It validates JWT tokens, sets by roles permissions, and secures all endpoints with access restriction.
- **Clinical Data Management Module:** It is in charge of the dependable and efficient persistent storage of clinical entities (patients, assessments, medications, activities, appointments) and retrieval using MongoDB.
- **Prediction Orchestration Module:** It checks and logs requests for predictions, sends them to the ML service, normalizes the replies, and stores the prediction results along with their metadata.
- **Audit Logging Module:** It is a recorder of all actions relating to security and clinical aspects which include authentication, data access, prediction requests, and exports.
- **Model Registry Module:** It keeps the metadata of machine learning models such as versioning, calibration parameters, and the status of the active deployment.

5.1.3 Machine Learning Service Modules

- Data Preprocessing Module: It performs the tasks of validating and normalizing the incoming multimodal inputs while at the same time maintaining the integrity of the subject-level split.
- Feature Encoding Module: It takes care of creating fixed-length embeddings for MRI, tabular cognitive data, and structured clinical inputs.
- Fusion Module: It is responsible for merging modality embeddings by using early or late fusion techniques, and at the same time being aware of the missing modality.
- Prediction Heads Module: This module gives the calibrated probabilities for the patient's current Alzheimer's status and future progression timelines (12–36 months) as the output.
- Calibration & Explainability Module: It does the probability calibration and also produces the explainability artifacts like Grad-CAM, SHAP values, and attention maps.

5.2 Interfaces Between Components

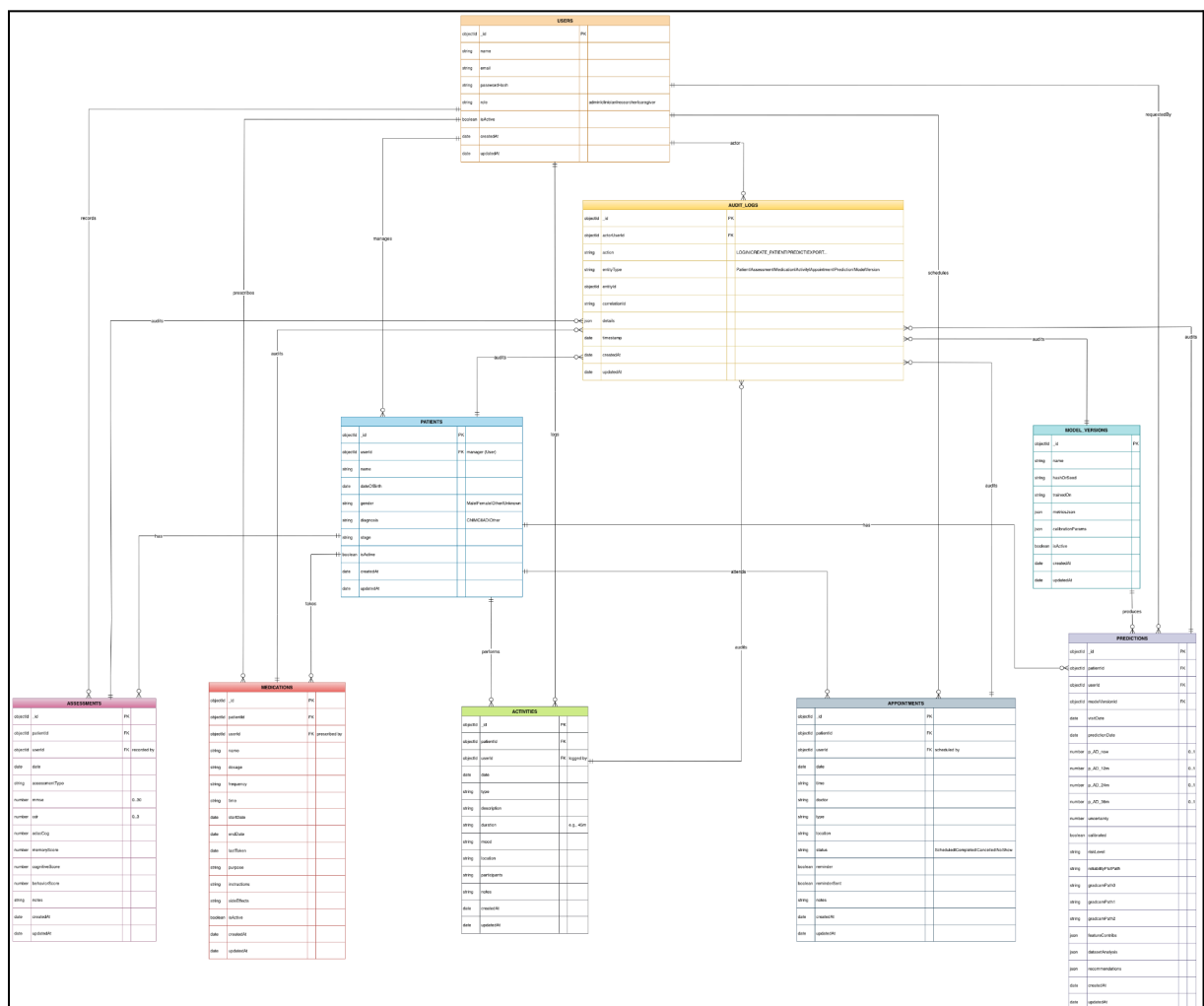
The system follows a layered interaction model:

- The Web user interface interacts solely with the Backend API through RESTful HTTP endpoints.
- The Backend API handles the permissions verification, request processing, and communication with the ML Service via secured REST interfaces.
- From the ML Service, the predicted outcomes and explainability artifacts stream to the Backend API, which then saves them and returns them to the UI.
- Asynchronous processing may be used for heavy explainability computations (such as Grad-CAM generation) to keep the UI responsive.
- Communication between all components includes metadata for traceability purposes, such as subject identifiers, visit context, and model version.

6 Data Design

In this part, a detailed account of the data structure, storage, and relationships within the system is given. The main persistence layer we prefer is MongoDB (Document store) with patient-centric collections and strict mediation through the Backend API. Big explainability artifacts like Grad-CAM images and reliability plots are stored as files/objects and referenced by path in MongoDB. All timestamps use the UTC ISO date format, and application models contain creation/update metadata for auditability purposes.

6.1 Database Schema



Storage Model and Relations:

- **Primary store:** MongoDB version 5.0 or later; one logical tenant for the project; the Patient is the aggregate root.
- **Core relations:** a Patient can be managed by a User; a Patient is the owner of Assessments, Medications, Activities, Appointments, and Predictions; a ModelVersion can create many Predictions; AuditLog keeps track of who did what on which entity and when.
- **Access pattern:** the Web UI does not communicate with MongoDB directly—the Backend API to which all reads/writes are done enforces RBAC, validation, and audit.

Collections:

- **Users:** The System accounts (Admin, Clinician, Researcher, Caregiver) are the users who are authenticated and authorized. A single user can manage or own several patients at a time and will thus be recorded as the person overseeing the clinical and operational changes.
- **Patients:** Patient's Demographics and clinical summary. It is the main source of all visit-level and longitudinal data. It can be traced back from every visit-level data collection through the patientId.
- **Assessments:** The cognitive/clinical measurements (e.g., MMSE, CDR-SB, ADAS-Cog) and notes are visit-level data. The patient and the user who recorded it are linked to this data.
- **Medications:** Prescriptions with information on dosage and frequency, start and stop dates, and adherence metadata. The patient and the user who prescribed or updated it are linked to this data.
- **Activities:** Daily-living or lifestyle events (type, date, duration, mood, notes). The patient and the user who logged it are linked to this activity data.
- **Appointments:** Scheduled visits (date/time, doctor, location, status, reminders). The patient and the user who scheduled or updated it are linked to this data.
- **Predictions:** Model outputs and context (probabilities for AD now and at 12/24/36 months, uncertainty, calibration flag, artifact references). The patient, the requesting user, and the model_versions entry used to generate the result are linked.
- **Model_versions:** Governance for deployed models (name, hash/seed, training split identifier, metrics, calibration parameters, active flag). One model version may generate many predictions.
- **Audit_logs:** Immutable trail of actions (actor user, action, entity type/id, details, correlation id, timestamp) covering authentication, CRUD, predictions, and exports.

Indexing and Performance:

- **Patient-centric queries:** compound indexes were made on visit-level collections by (patientId, date) for speedy timelines.

- **Directory/search:** a unique index was created on users.email; text or case-insensitive search index was created on patients.name; filters were applied on diagnosis/status/active flags.
- **Governance/audit:** the model_versions.isActive field is quickly referenced; time-range and actor/entity filters can be applied on audit_logs.
- **Artifacts:** only file/URL references are stored in MongoDB; binary assets are in file/object storage.

6.2 Data Dictionary

- **Users:** The fields consist of name, email (which is unique and should be in lowercase), password hash, role (admin/clinician/researcher/caregiver), an active flag, and timestamps. They are used for authentication, RBAC, and as the actor recorded in audit logs and provenance fields across the domain.
- **Patients:** The core identity and clinical summary include full name, date of birth, gender, diagnosis (CN/MCI/AD/Other), optional stage, optional baseline indicators (like education years, APOE-ε4 count, latest MMSE/CDR), an active flag, manager userId, and timestamps. It is the parent in composition relationships to all visit-level data.
- **Assessments:** Visit metadata including date and assessment type, recorder userId, and cognitive measures as well as free-text notes (MMSE 0–30, CDR 0–3, ADAS-Cog and optional subscores). Semantics: each assessment is linked to exactly one patient and one recording user; they are sorted by date for longitudinal views.
- **Medications:** Information about drug name, dosage, frequency/time, start/end dates, last-taken marker, purpose/instructions/side-effects, active flag, patient link, and actor userId. Semantics: facilitates adherence and medication history for each patient.
- **Activities:** Information about date, type, description, duration, mood, and optional location/participants/notes, along with patient link and actor userId. Semantics: adds lifestyle context to clinical measures.
- **Appointments:** Information about date/time, doctor, type, location, status (Scheduled/Completed/Cancelled/NoShow), reminder flags, notes, and patient link plus scheduler userId. Semantics: aids in care coordination and managing upcoming visits.
- **Predictions:** The context fields give the visit date and the date when the prediction was made, three links to the patient, the user, and the model version are also part of the predictions. The outputs consist of calibrated probabilities ($p(\text{AD_now})$, $p(\text{AD_12m})$, $p(\text{AD_24m})$, $p(\text{AD_36m})$), an indication of uncertainty, a risk label that is optional, and references to artifacts (paths to the reliability plot, Grad-CAM images, and tabular/text feature-contribution in JSON format).
- **Model Versions:** The information that is stored includes the name, reproducibility identifier (which is either a hash or seed), training/split descriptor, metrics (like

AUC/PR-AUC, ECE/Brier), calibration parameters, active flag, and timestamps.

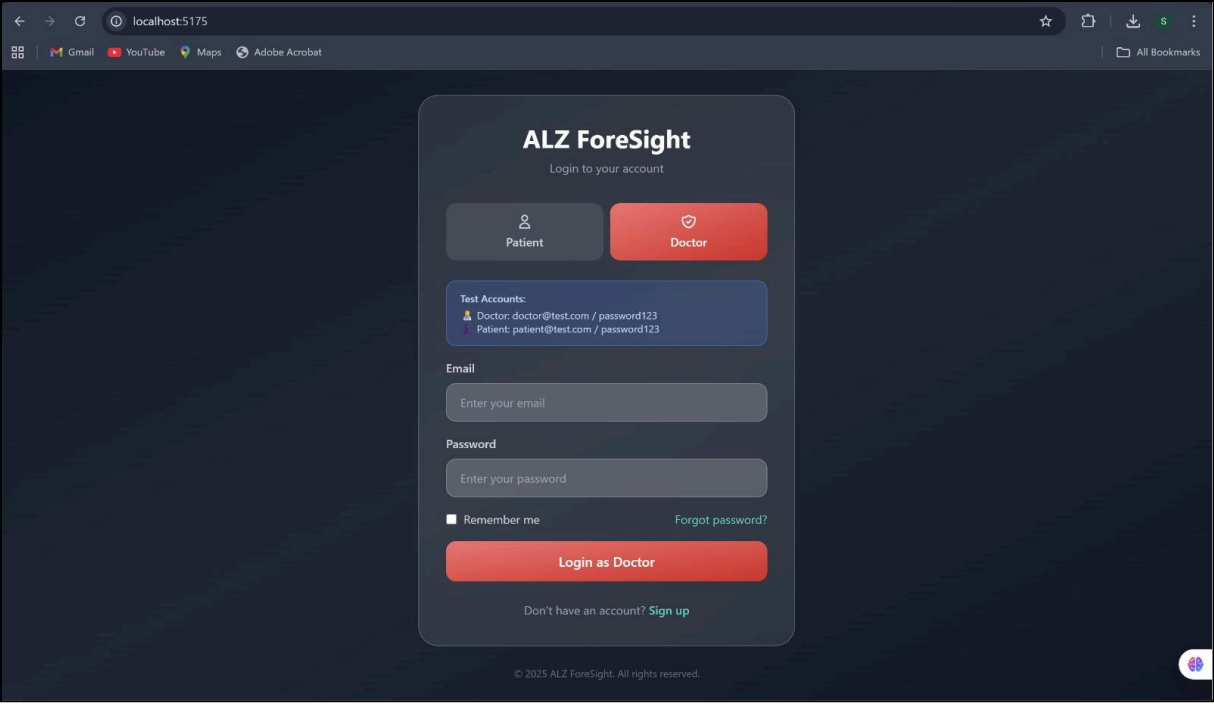
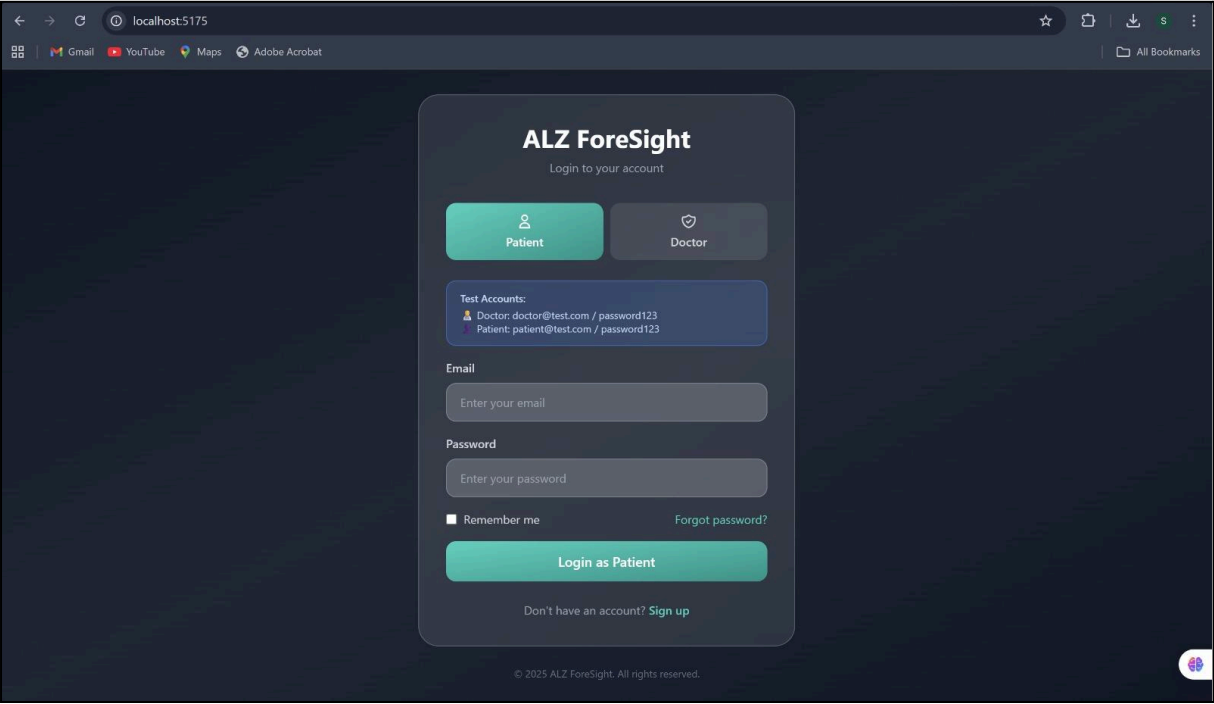
Semantics: allows for very strict traceability from any prediction back to the exact model and calibration in use.

- **Audit Logs:** The actor's user ID, the action verb (for example, LOGIN, CREATE_PATIENT, PREDICT, EXPORT), the entity type and id, an optional details payload, a correlation id for request tracing, and a timestamp. Semantics: makes the issue of accountability and support for grading/review more robust; it is an append-only log and is never edited.

7 User Interface Design

The screenshot shows a web browser window with the URL `localhost:5175/ai-diagnosis`. The application is titled "AI-Powered Alzheimer's Diagnosis" and uses an "Advanced XGBoost model for cognitive assessment". It features three tabs: "Basic Info", "Cognitive Tests", and "Brain Imaging". The "Basic Info" tab is active, showing input fields for "Age (years) *" (with a placeholder "e.g., 75.5"), "Gender" (a dropdown menu with "Male" selected), "Education (years)" (with a placeholder "e.g., 16"), and "APOE4 Alleles" (a dropdown menu with "0 (No risk)" selected). A green "Get AI Diagnosis" button is positioned below these fields. To the right of the input fields is a large, dimmed area with a lightbulb icon and the text "Enter patient data to get AI diagnosis" and "Results will appear here". A "Quick Guide" section is located at the bottom left, providing instructions: "Fill in Basic Info, Cognitive Tests, and Brain Imaging tabs", "Required fields marked with *", "More data = better prediction accuracy", and "Optional fields can be left blank".

The screenshot displays the "ALZ ForeSight" patient dashboard. At the top, there is a search bar and navigation icons. The main section is titled "Overview" and "Patient Health". It features a profile card for "Jane Cardiologist" with a "Video call" button. Below this is a 3D brain model with a highlighted area, labeled "Brain Activity 95% Active". A circular progress indicator shows "5" and the time "6:41 PM" on "Feb 2023". To the right is a calendar for "Feb 2023" with the 15th highlighted. Below the calendar is an "In-Week Visits" section with three cards: "26 April 23 Complete Blood Count (CBC) Dr. Alberto Therapist", "1 May 23 Clinic Visit Appointment Dr. Lucia Rdc", and "02 June 23 Video Consultation Chat Dr. Mark Aryan".



localhost:5175/ai-diagnosis

Gmail

YouTube

Maps

Adobe Acrobat

All Bookmarks



AI-Powered Alzheimer's Diagnosis
Advanced XGBoost model for cognitive assessment

Basic Info

Cognitive Tests

Brain Imaging

MMSE (0-30) *

ADASI3

e.g., 24

e.g., 18.5

ADAS11

CDR-SB

Optional

Optional

MOCA

FAQ Score

Optional

Optional

RAVLT Immediate

RAVLT Learning

Optional

Optional

Get AI Diagnosis

Quick Guide

- Fill in **Basic Info**, **Cognitive Tests**, and **Brain Imaging** tabs
- Required fields marked with *
- More data = better prediction accuracy
- Optional fields can be left blank



Enter patient data to get AI diagnosis
Results will appear here

localhost:5175/ai-diagnosis

Gmail

YouTube

Maps

Adobe Acrobat

All Bookmarks



AI-Powered Alzheimer's Diagnosis
Advanced XGBoost model for cognitive assessment

Basic Info

Cognitive Tests

Brain Imaging

Hippocampus Volume (mm³) *

e.g., 3500.5

Ventricles (mm³)

Whole Brain (mm³)

Optional

Optional

Entorhinal (mm³)

Fusiform (mm³)

Optional

Optional

MidTemp (mm³)

ICV (mm³)

Optional

Optional

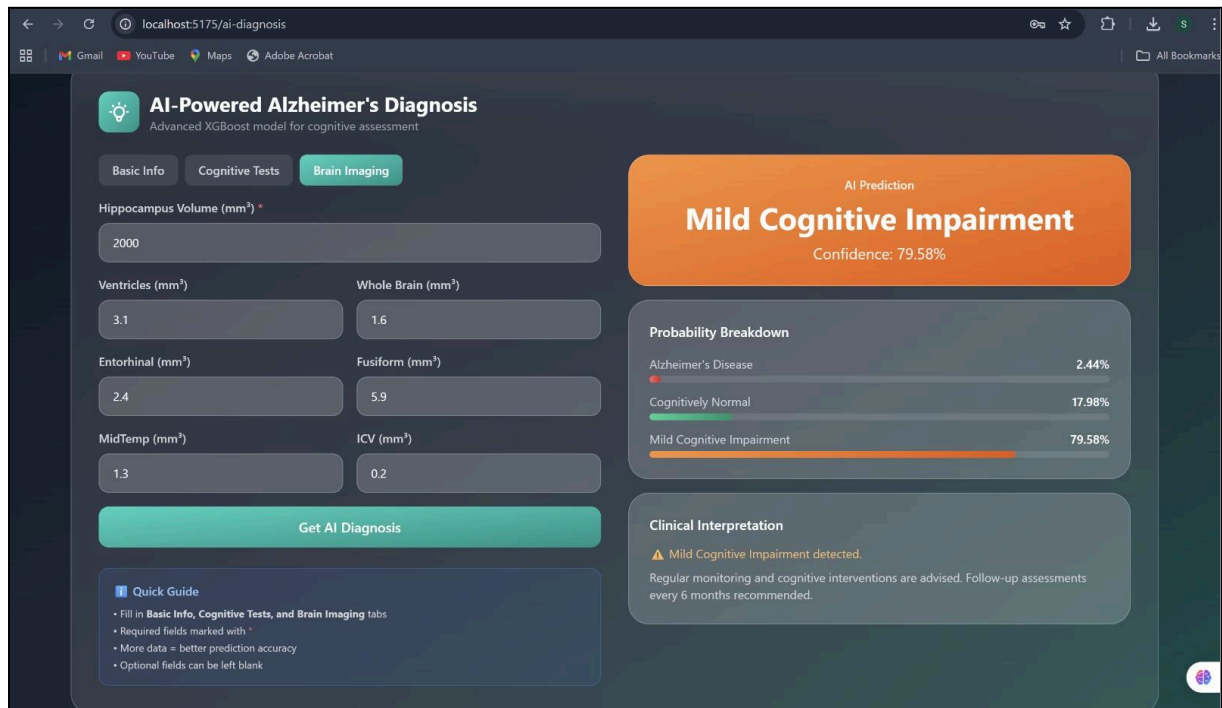
Get AI Diagnosis

Quick Guide

- Fill in **Basic Info**, **Cognitive Tests**, and **Brain Imaging** tabs
- Required fields marked with *
- More data = better prediction accuracy
- Optional fields can be left blank



Enter patient data to get AI diagnosis
Results will appear here



7.1 Screen Prototypes or Wireframes

- Sign-In: Single sign-on for email and password, research disclaimer, error feedback; routing to home page based on user role.
- Global Shell: Left-hand vertical menu (Dashboard, Patients, Assessments, Medications, Activities, Appointments, Predictions, Exports; Admin: Users, Models, Audit). Search, model badge, and user menu are available on the top bar.
- Patient List: Search/filter, add patient, rows/cards with quick actions (View, Assess, Predict) available.
- Patient Profile: Top (demographics, diagnosis, latest risk) and panels: Risk & Uncertainty (gauges), Trends (MMSE/CDR/ADAS-Cog), Upcoming Care (appointments, meds).
- Assessments / Medications / Activities / Appointments: Clean tables or timelines with "Add / Edit / Delete," validated forms, toasts, and audit.
- Prediction (Enhanced) Result: Calibrated probabilities for now/12/24/36m, uncertainty note, explainability panels (MRI Grad-CAM thumbnails; feature contributions), Save/Export actions.
- Admin: User & Roles (invite, activate/deactivate), Model Registry (set active, view metrics), Audit Logs (filter by actor/entity/time).

7.2 User Interaction Flow

- Authenticate: Signing in leads to a role-based landing page (Clinician: Patients; Researcher: Predictions; Admin: Users/Models).
- Find Patient: Search/filter → Profile opens → risk snapshot and trends take place in review.
- Record Data: Validation form for the assessment is added, and this is followed by the refresh of the charts and the writing of the audit entry.
- Run Prediction: Press “Predict (Enhanced)” → API calls ML → display calibrated risks + explanations → Save/Export.
- Manage Care: Medications and Appointments are updated → adherence/status appears on Profile.
- Admin/Governance: users/roles management; switching active model; audit logs review.
- Resilience: Validation errors or ML downtime are communicated clearly; placeholders are shown for long jobs; saved data and audits are retained.

8 External Interfaces

8.1 API and System Interfaces

8.1.1 Backend REST API

Backend API provides RESTful endpoints to utilize system features among others:

- Authentication APIs: /api/auth/login, /api/auth/logout, /api/auth/refresh
- Clinical Data APIs: /api/patients, /api/assessments, /api/medications, /api/activities, /api/appointments
- Prediction APIs: /api/predictions/predict/api/predictions/predict-enhanced
- System Health & Metadata APIs: /api/health, /api/models, /api/audit

All endpoints require the use of JWT authentication as well as role-based access control measures for security purposes.

8.1.2 Machine Learning Service API

ML service works separately as a Python-based REST service and opens:

- GET /health – Check service status
- POST /predict – Prediction of baseline
- POST /predict-enhanced – Prediction with understanding
- GET /model-info – Metadata of current model

Backend API and ML service communicate through HTTP with timeouts, retries, and response formats enforced as per standard.

8.1.3 Database Interface

MongoDB serves as the main database for clinical records, predictions, audit logs, and model metadata, among others. All the database accesses are done through Backend API; UI or ML service could not access the database directly.

8.1.4 External Constraints

There is no direct integration with hospital EHR systems (e.g., HL7/FHIR) included in the project scope. Data utilization is in accordance with the ADNI data use agreements throughout. Only de-identified and approved research data are being stored and provided for further usage.

9 Traceability Matrix

Req ID	What it is	Where it lives	How we check it
FR-01.2 Feature extraction	ML encoders for MRI/Text/Tabular	ML Service (encoders), API /predict	Unit tests + E2E demo case
FR-01.3 Fusion & missing data	Early/late fusion, modality mask	ML fusion module; API schema	Ablation runs + E2E
FR-01.4 Two-stage prediction	Now + 12/24/36m heads	ML heads; UI risk gauges	API schema + UI shows all horizons
FR-01.5 Calibration & XAI	Temp/Platt; Grad-CAM/feature contribs	ML post-hoc; UI explain panels	ECE/Brier targets + visual check
FR-02.1 Auth/RBAC	JWT + roles	API middleware; UI protected routes	Auth tests + E2E
FR-02.2 Clinical CRUD	Patients, assessments, meds, etc.	API resources; Mongo	CRUD tests + audit entries
FR-02.3 Prediction bridge	API→ML proxy, health checks	API ML client; /health	Fault-injection + E2E
FR-03.1 Patient dashboard	Gauges, trends, adherence	UI dashboard; summary APIs	Loads fast; values match DB
FR-03.2 Explainability views	Saliency/feature panels	UI viewers; artifact links	Renders per prediction
FR-05.1 Audit trails	Who did what/when	API middleware; audit_logs	Every key action logged
FR-05.2 Model versioning	Register/set active version	/models registry; metadata	Active model stored with results

FR-05.3 Errors & alerts	Friendly errors; health status	API/ML error mappers; toasts	Negative tests pass
FR-05.4 Export	CSV/PDF (anonymized)	/exports + UI button	No PHI; columns correct
FR-06.1 Consent/policy	Enforce ADNI policy	API guards; UI badges	Blocked fields never leak
FR-06.2 Education	Disclaimers & thresholds info	UI info modals	Visible before use
NF-R1 Security	RBAC, rate limits, CORS/CSRF	API gateway	Baseline pen-test
NF-R2 Reliability	Health, retry, backup	API/ML, ops guide	Health green; restore works
NF-R3 Maintainability	Modular code + CI tests	Repo/CI	80%+ core coverage
NF-R4 Portability	Docker/Compose	Root compose	Runs locally easily
NF-R5 Extensibility	Pluggable models	Encoder/registry	Swap model w/o UI change
NF-R6 Accessibility	Basic a11y	UI components	Keyboard/alt text OK
NF-R7 Performance	Bounded inference	ML settings	CPU demo < target secs
NF-R8 Serviceability	Diagnostics + error IDs	Admin page	Faster triage
NF-R9 Activity logging	Full audit on CRUD/predict/export	Middleware	100% coverage of key ops

9.1 Requirements-to-Design Mapping

Requirement	Design element
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FR-01.2 Feature extraction	ML encoders + standardized feature schemas
FR-01.3 Fusion & missing data	Configurable fusion_strategy + modality_mask
FR-01.4 Two-stage prediction	Separate heads (now / 12 / 24 / 36m)
FR-01.5 Calibration & XAI	Post-hoc calibration + Grad-CAM/feature contrib artifacts
FR-02.1 Auth/RBAC	JWT middleware; role-guarded routes
FR-02.2 Clinical CRUD	REST resources per entity; patient-centric indexes
FR-02.3 Prediction bridge	API→ML proxy with timeout/retry; /health
FR-03.1 Dashboard	Summary endpoints + UI dashboard cards/charts
FR-03.2 Explainability views	Artifact URLs + JSON attributions rendered in UI
FR-05.1 Audit trails	Audit middleware + audit_logs collection
FR-05.2 Model versioning	/models registry + active flag stored with results
FR-05.3 Errors & alerts	Unified error format + toasts + health flags
FR-05.4 Export	/exports CSV/PDF with anonymization
FR-06.1 Consent/policy	Field blockers + consent flags at API layer
FR-06.2 Education	In-app disclaimers and threshold/uncertainty notes
NF-R1 Security	Rate limit, CORS/CSRF, HTTPS in prod
NF-R2 Reliability	Health probes, retries, backups/restore
NF-R3 Maintainability	Layered services, lint/tests, CI
NF-R4 Portability	Docker/Compose + env vars
NF-R5 Extensibility	Encoder/fusion registry; model registry
NF-R6 Accessibility	WCAG-aware components
NF-R7 Performance	Bounded batch, CPU fallback
NF-R8 Serviceability	Diagnostics page + error IDs

NF-R9 Activity logging	Middleware capturing CRUD/predict/export
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10 Design Constraints

10.1 Technical / Tooling Constraints

- **Stack lock-in:** Existing codebase and team skillset constrain us to **React (Web UI)**, **Node.js/Express (API)**, **Python (Flask/FastAPI for ML)**, and **MongoDB**. Major rewrites to other stacks (e.g., Java/.NET, SQL RDBMS) are out of scope.
- **Version pinning:** Frameworks, CUDA/cuDNN, PyTorch/Sklearn, Node LTS, and MongoDB are **pinned** to avoid drift; upgrading mid-semester is risky and limited to security patches.
- **Library licensing:** Only OSS licenses compatible with academic use are permitted; some desired medical-imaging toolkits may be excluded due to licensing.
- **Model formats:** Inference paths assume **PyTorch** or **ONNX** artifacts; TensorFlow-only models are not first-class.

10.2 Platform / OS Constraints

- **Target OS:** Development on **Windows/macOS/Linux**; deployment on **Linux x86_64** servers/VMs or containers.
- **Containerization:** Runtime assumes **Docker/Compose**; bare-metal installs are supported only for local dev.
- **GPU drivers:** GPU builds require compatible **NVIDIA drivers** and **CUDA** versions; mismatches will force CPU fallback.

10.3 Hardware Constraints

- **GPU availability:** MRI inference/explainability (e.g., Grad-CAM) is **GPU-preferred**; when unavailable, CPU paths are enabled but slower and may downscale images.
- **Memory & storage:** MRI volumes and embeddings are large; minimum suggested **16–32 GB RAM** for dev machines, and **fast SSD** for MongoDB and artifact caching.
- **Throughput limits:** Batch sizes for inference are capped to prevent OOM on student hardware; background jobs throttle to avoid contention.

10.4 Data & Compliance Constraints

- **ADNI policy:** Only **de-identified, policy-compliant** data may be used; redistribution is

prohibited. Product ships with **synthetic/demo** data for evaluators.

- **PHI & re-identification:** Fields that can re-identify subjects are **blocked**; exports are anonymized by design.
- **Subject-level splits:** All training/validation/test processes must honor **subject-level** separation to avoid leakage; this constrains how data loaders and CV are implemented.

10.5 Network & Environmental Constraints

- **Latency / bandwidth:** MRI artifact downloads can be large; UI requests may use **lazy loading** and **progressive rendering**. Slow links will impact saliency/plot load times.
- **Offline operation:** Full offline use is **not supported**; the UI requires the API, and the API requires the ML service and DB.
- **Asynchrony:** Heavy explainability (e.g., Grad-CAM) may run **asynchronously**; users may see placeholders until artifacts are ready.

10.6 Deployment & Ops Constraints

- **Single-VM baseline:** Grading/demo assumes a **single-VM or compose** deployment; no Kubernetes required. Horizontal scaling is optional and limited by hardware.
- **Secrets management:** For coursework, **.env files** are used; enterprise vaults are out of scope.
- **Backups & DR:** Mongo backups are **manual or scheduled** scripts; no managed database SLA.

10.7 Browser / UX Constraints

- **Browser support:** The latest versions of Chrome, Edge, and Firefox are supported; Safari support is best-effort. Older browsers (IE/old Android WebView) are not included.
- **Accessibility:** Only basic WCAG practices are implemented (accessibility via keyboard, alt texts, contrasting colors); there is no guarantee of full screen-reader parity.
- **Localization:** UI will be available only in English for this phase; i18n hooks are optional.

10.8 Integration Constraints

- **No EHR/HL7/FHIR:** This project does not include direct hospital system integrations, which are out of scope.
- **Email/SMS:** Notifications are optional and depend on third-party services; demo environments may have them disabled.

10.9 Testing & Reproducibility Constraints

- Compute limits: The entire MRI training cannot be done within the course resources; rather, the focus will be on inference, calibration, and ablations using small samples.
- Determinism: The exact reproduction of results is only possible when using fixed seeds, pinned libraries, and stable hardware; however, there might be minor numeric drifts across different platforms/GPUs.

11 Glossary

1. AD (Alzheimer's Disease): A gradually advancing neurodegenerative illness that has memory and cognitive functions as its main victims.
2. ADAS-cog (Alzheimer's Disease Assessment Scale–Cognitive Subscale): A widely accepted and reliable test to assess cognitive abilities; usually a table format is used as an input.
3. ADNI (Alzheimer's Disease Neuroimaging Initiative): A cohort to conduct research for a long term and provide MRI and clinical data under very strict data-use policies.
4. Artifact (ML/System): The output or file that is produced by the system (e.g., model weights, calibration files, Grad-CAM images, reliability plots, exports, and audit logs).
5. AUC / AUROC (Area Under the ROC Curve): A discrimination metric that tells how good a model is at separating the classes across the thresholds.
6. Audit Log: A record that cannot be altered and shows who did what, when, and to which entity.
7. Baseline (Visit): The first visit of the patient that is taken as a reference point for the longitudinal analysis.
8. Brier Score: A metric for calibration that evaluates the mean squared error of the probabilities predicted against the actual outcomes.
9. Calibration (Probability Calibration): A post-hoc adjustment (e.g., temperature scaling, Platt scaling) that makes the predicted probabilities in line with the observed frequencies.
10. Caregiver/Patient View: A less complicated dashboard that presents the main risks, explanations, and suggestions in a simplified manner.
11. CDR-SB (Clinical Dementia Rating – Sum of Boxes): The clinical staging measure used as a tabular input and outcome context.
12. Class Diagram: A structural model depicting the classes/entities, attributes, methods,

and relationships.

13. CRUD (Create, Read, Update, Delete): The basic operations for record management via both the application and the API.
14. De-identification / PHI: The process of eliminating personally identifiable health information; only the approved research fields are stored/exported.
15. Early Fusion: Merging modality features at the feature level prior to the prediction heads.
16. Explainability / XAI: Techniques that assist in understanding model predictions (e.g., Grad-CAM, SHAP, attention).
17. Feature Extraction / Embedding: The process of converting raw data (MRI, text, tabular) into fixed-length numerical vectors suitable for modeling.
18. Fusion Strategy: The pre-defined method (early or late fusion) for the combination of modalities, with rules for handling missing data.
19. Grad-CAM (Gradient-weighted Class Activation Mapping): The technique of saliency that marks the MRI regions that greatly affect the prediction.
20. Harmonization (Imaging): Procedures that help eliminate the variations in MRI data caused by a specific site, scanner, or protocol.
21. Health/Readiness Endpoint: A service URL that indicates live status and readiness (e.g., /health).
22. Horizon: A future period of time (e.g., 12, 24, 36 months) for which the risk of progression is predicted.
23. JWT (JSON Web Token): A token that is signed and used for user authentication and carrying role/claim information.
24. Late Fusion: Prediction of all modalities is done at the decision level with combining methods like averaging/stacking of probabilities.
25. Leakage (Data Leakage): Unintentional flow of information between the train/validation/test phases (e.g., visits of the same subject are split across folds).
26. MCI (Mild Cognitive Impairment): The stage of a person being cognitively normal but

very close to Alzheimer's disease.

27. MMSE (Mini-Mental State Examination): A cognitive test that ranges from 0 to 30, often considered as a tabular feature.
28. Model Registry / Model Version: A database that stores the metadata related to models (name, hash/seed, metrics, calibration parameters), and the currently active version.
29. Modality: A specific type or category of data source (e.g., an MRI image, clinical nurse's notes, tabular features).
30. Modality Dropout: A training/serving technique that is used to maintain resilience in case one or more modalities are not available.
31. MongoDB: A document-oriented database that serves for storing users, patients, clinical records, predictions, models, and audits.
32. Patient Aggregate (Aggregate Root): A concept in domain design where the Patient takes possession/parent role of visit-level records.
33. Platt Scaling: A method for calibrating probabilities based on logistic regression.
34. PR-AUC (Area Under the Precision–Recall Curve): A metric that focuses on the performance of the positive class.
35. Presence Mask: A Boolean variable that indicates the modalities available for a certain case.
36. Prediction Heads: The layers at the output that give probabilities for the current AD status and future horizons (12/24/36 months).
37. Provenance: The ability to trace back actions/data to their source (who/when/which version/config).
38. RBAC (Role-Based Access Control): A method of authorization that groups user roles (Admin, Clinician, Researcher, Caregiver) and limits their actions accordingly.
39. Reliability Diagram: A graphic that helps to assess calibration by comparing predicted probabilities to observed frequencies.
40. REST API: An HTTP interface that makes resources available for the UI and other

clients (e.g., /patients, /predict).

41. Reproducibility (ML): The process of obtaining the same results again and again by fixing seeds, versions, splits, and documenting configurations/artifacts.
42. Seed / Hash (Reproducibility ID): A numerical seed or a content hash that is applied for the subject/model training and evaluation to be reconstituted.
43. SHAP (Shapley Additive explanations): A method that attributes inputs by giving a numerical value to each input according to its contribution to a particular prediction.
44. Subject-level Split: A rule of cross-validation that prevents leakage by keeping all of a subject's visits in the same fold.
45. Temperature Scaling: A method of calibration that involves the rescaling of logits to better align probabilities.
46. T1-MRI: A very detailed structural MRI sequence that is applied to volumetric and morphometric analysis.
47. Two-Stage Prediction: Initially determine present AD condition, after that predict the risk of progression at later times.
48. Uncertainty (Predictive): A single value or a distribution representing the level of certainty in the prediction (for instance, variance or entropy).
49. Use Case Diagram: A model of behavior displaying the stakeholders, their intended outcomes (use cases), and relationships of inclusion/extension.
50. Web Dashboard: The interface for doctors/researchers showing the risks, uncertainty bands, explanations, trends over time, and tools for management.

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